Task 4: Classification with Logistic Regression

Important output screenshots from the executed Google Colab notebook

Dataset: Breast Cancer Wisconsin (Diagnostic) Dataset

Objective: Build and evaluate a binary classification model using Logistic Regression.

Tools: Python, Pandas, Matplotlib and Scikit-learn

This PDF collects the key notebook outputs: dataset preview, data split, threshold selection, evaluation metrics, confusion matrix, ROC curve and sigmoid visualization.

1. Dataset Preview and Class Distribution

```
Dataset loaded successfully!
Dataset shape: (569, 31)

Class distribution:
diagnosis
benign       357
malignant    212
Name: count, dtype: int64

Dataset preview:
   mean radius  mean texture  ...  mean smoothness  diagnosis
0        17.99         10.38  ...          0.11840  malignant
1        20.57         17.77  ...          0.08474  malignant
2        19.69         21.25  ...          0.10960  malignant
3        11.42         20.38  ...          0.14250  malignant
4        20.29         14.34  ...          0.10030  malignant

[5 rows x 6 columns]
```

2. Training and Testing Data Split

```
Training data: (455, 30)
Testing data: (114, 30)

Training class counts:
target
Benign       285
Malignant    170
Name: count, dtype: int64

Testing class counts:
target
Benign       72
Malignant    42
Name: count, dtype: int64
```

3. Feature Standardization

```
Features standardized successfully!
Fitting data: (364, 30)
Validation data: (91, 30)
```

4. Threshold Tuning Results

```
Threshold comparison:
    Threshold  Precision  Recall  F1 Score
0        0.20      0.825   0.971     0.892
1        0.25      0.868   0.971     0.917
2        0.30      0.943   0.971     0.957
3        0.35      0.943   0.971     0.957
4        0.40      0.943   0.971     0.957
5        0.45      0.943   0.971     0.957
6        0.50      0.943   0.971     0.957
7        0.55      0.943   0.971     0.957
8        0.60      0.943   0.971     0.957
9        0.65      1.000   0.971     0.985
10       0.70      1.000   0.971     0.985
11       0.75      1.000   0.971     0.985
12       0.80      1.000   0.971     0.985Selected threshold: 0.65
The threshold is selected using validation data, not the test data.
```

5. Final Model Evaluation

```
Selected threshold: 0.65
Confusion Matrix:
[[72  0]
 [ 4 38]]

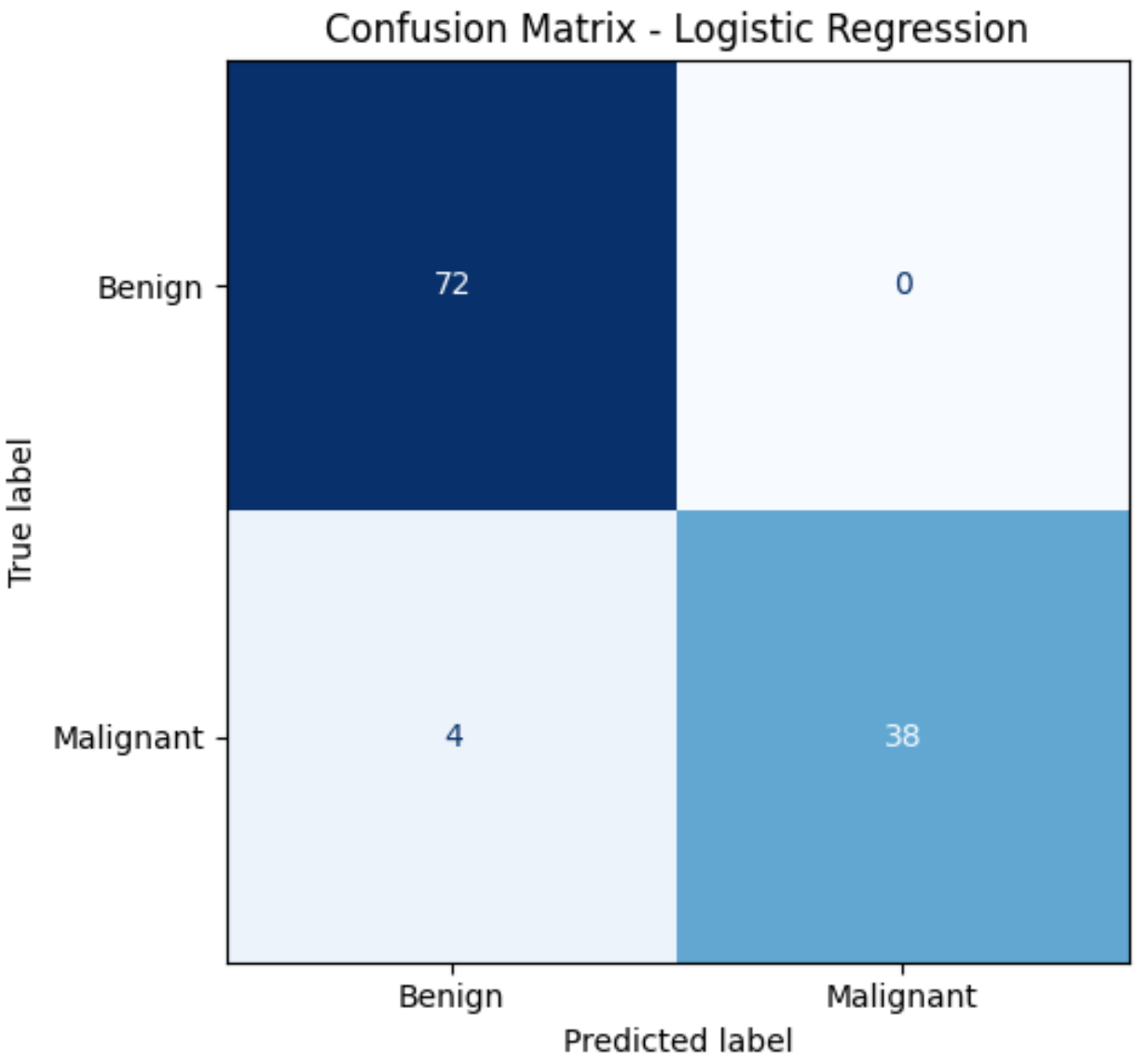
Precision (Malignant): 1.0
Recall (Malignant): 0.9048
ROC-AUC Score: 0.996

Classification Report:
              precision    recall  f1-score   support

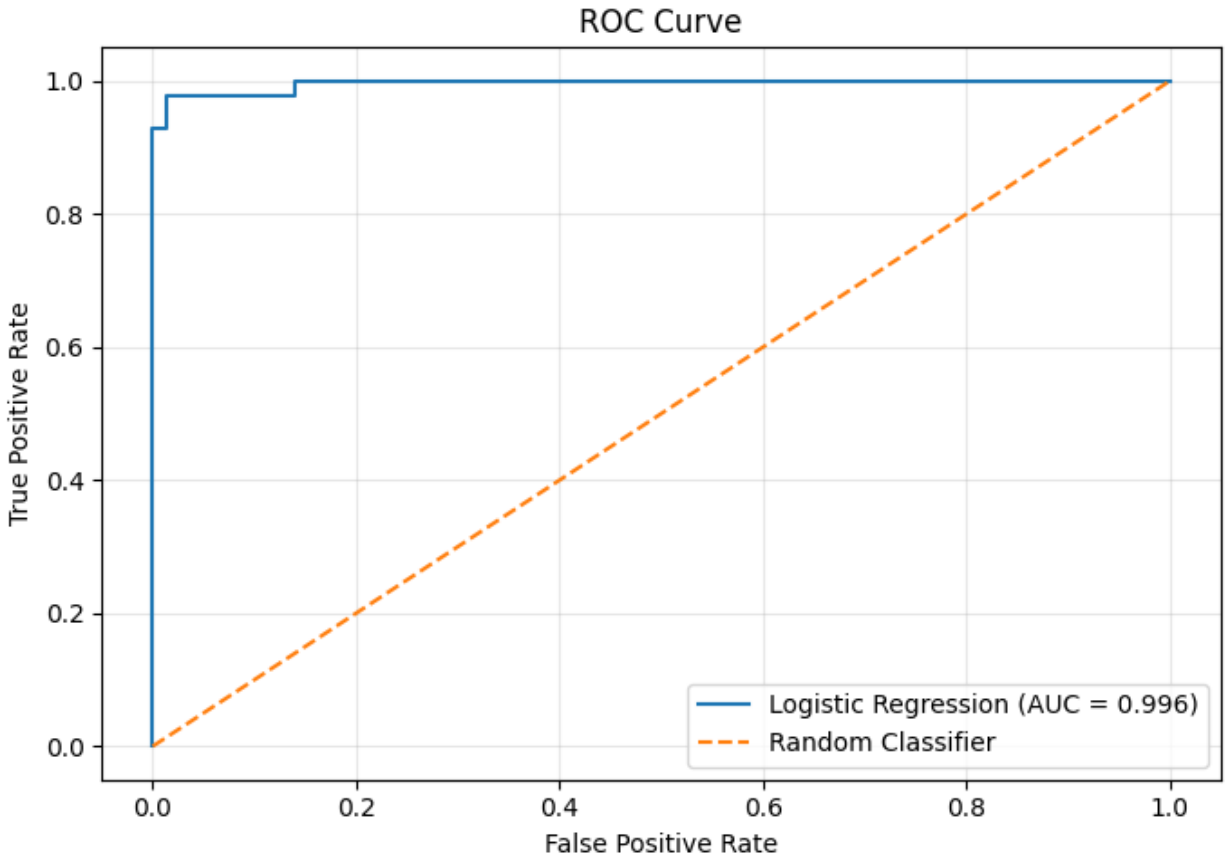
      Benign       0.95      1.00      0.97        72
   Malignant       1.00      0.90      0.95        42

    accuracy                           0.96       114
   macro avg       0.97      0.95      0.96       114
weighted avg       0.97      0.96      0.96       114
```

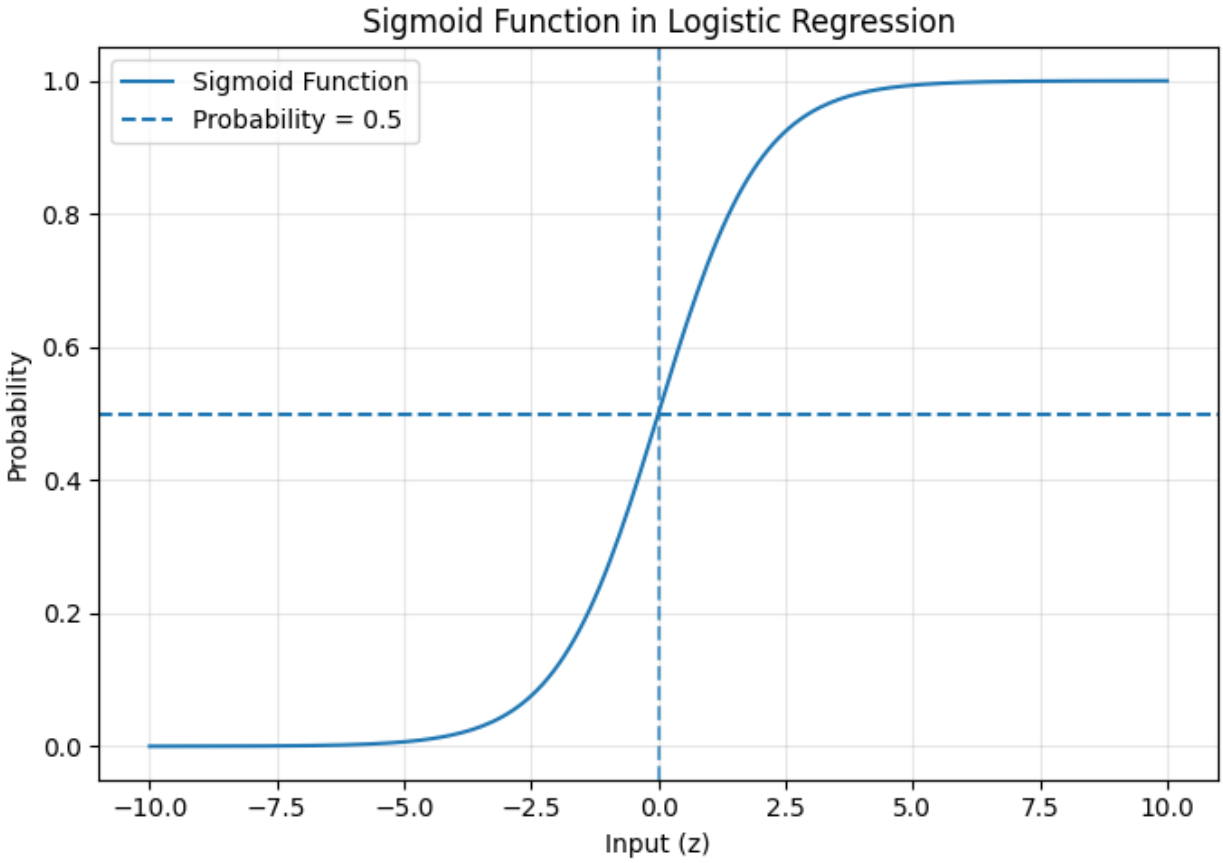
6. Confusion Matrix



7. ROC Curve


ROC Curve
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Logistic Regression (AUC = 0.996)
Random Classifier

8. Sigmoid Function



9. Precision, Recall and F1 Score by Threshold

